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Subject : Fundamentals of Telecommunication Systems

Experiment Number : 8

Experiment Title : Demonstration of Signal Transmission through Wireless Media (Wi-Fi)

Date of Performance : 23th March 2026

Date of Submission : 23th March 2026

Experiment No. 08: Demonstration of Signal Transmission through Wireless Media (Wi-Fi)

Theory

1. What is Wireless Media?

Wireless media is a type of transmission medium that allows data, voice, and video to be transmitted without the use of physical cables or wires. Instead of copper wires or fiber-optic cables, wireless media uses electromagnetic waves to carry signals through the air or space.

Types of wireless media include:

- Radio waves – used in Wi-Fi, Bluetooth, and radio communication
- Microwaves – used in satellite communication and mobile networks
- Infrared waves – used in remote controls and short-range communication

Advantages of wireless media:

- Provides mobility and flexibility
- Easy and quick to install
- Supports communication in hard-to-reach areas

Disadvantages:

- More prone to interference
- Generally less secure than wired media
- Performance depends on signal quality

2. Define SSID (Service Set Identifier) and its Purpose in a Wi-Fi Network

An SSID (Service Set Identifier) is a unique identifier or name assigned to a wireless local area network (WLAN). It is broadcast by a wireless access point or router so that nearby devices can detect and connect to the network.

Purpose of SSID:

- Identifies a specific wireless network among multiple networks
- Allows users to select the correct network to connect to
- Helps manage network access and security policies
- Supports multiple networks operating in the same area without confusion

Key points:

- Maximum length: 32 characters
- Can be visible (broadcasted) or hidden
- Multiple access points can share the same SSID to extend coverage

3. Difference Between a Wired LAN and a Wireless LAN (WLAN)

A Local Area Network (LAN) connects devices within a limited area such as a home, office, or campus. Based on the transmission method, it can be wired or wireless.

Basis of Comparison	**Wired LAN**	**Wireless LAN (WLAN)**
Transmission Medium	Uses physical cables (Ethernet, fiber optic)	Uses radio waves
Connection Type	Physical, cable-based connection	Wireless connection
Mobility	Very limited	High mobility
Installation	Complex and time-consuming	Easy and quick
Speed	Generally higher and more stable	Lower than wired LAN
Reliability	Very reliable	Affected by interference
Interference	Minimal	High (walls, devices, signals)
Security	More secure	Requires encryption and authentication
Basis of		

Comparison	Wired LAN	Wireless LAN (WLAN)
Cost	Higher initial setup cost	Lower installation cost
Maintenance	Difficult	Easier
Flexibility	Less flexible	Highly flexible
Examples	Office Ethernet network	Wi-Fi network

4. Significance of Signal Strength in Wireless Transmission

Signal strength refers to the power level of the wireless signal received by a device. It plays a crucial role in determining the quality, speed, and reliability of wireless communication.

Importance of signal strength:

- Strong signals result in faster data transfer rates
- Weak signals cause slow speed, packet loss, and disconnections
- Determines the effective coverage area of a wireless network

Factors affecting signal strength:

- Distance between transmitter and receiver
- Physical obstacles (walls, buildings, furniture)
- Interference from other wireless devices
- Antenna type and transmission power

Why it matters:

- Ensures stable communication
- Improves user experience
- Supports efficient network performance

Hardware Components Required

- **Wireless Router (WRT300N):** The central device that broadcasts the wireless signal and manages the network.
- **Laptop-PT (x2):** End devices equipped with wireless network interface cards.
- **Smartphone (x1):** A mobile device with built-in Wi-Fi capability.

Hardware and Connection Setup

- **Wireless Standard:** 2.4 GHz 802.11n
- **Network ID (DHCP):** 192.168.0.0 (Default for WRT300N)
- **SSID (Network Name):** MyWirelessNetwork
- **Authentication:** WPA2-Personal

Procedure

Step 1: Physical Topology Design

- Place one **Wireless Router (WRT300N)** in the center.
- Place two **Laptops** and one **Smartphone** near the router.
- Naming: Rename Smartphone following the convention (e.g., S11_01).
- **Observation:** Note that the Smartphone connects automatically (default settings), but the Laptops do not.

Step 2: Hardware Identification (Laptop Modification)

- Click on a **Laptop** > **Physical** tab.
- **Task:** Turn off the Laptop power. Remove the Ethernet module (RJ45) and drag the **WPC300N** (Wireless Module) into the slot. Turn the power back on.
- **Observation:** A dashed line appears between the Laptop and Router, indicating a wireless link.

Step 3: Wireless Router Configuration

- Click the **Wireless Router** > **Config** tab > **Wireless**.
- Set the **SSID** to MyWirelessNetwork.
- Select **WPA2-PSK** and enter a Passphrase (e.g., TSEC@2026).
- **Observation:** The dashed lines disappear because the clients now need the correct security key to connect.

Step 4: Connecting Clients to the Wireless Signal

- Click **Laptop** > **Desktop** tab > **PC Wireless**.
- Go to the **Connect** tab. Click **Refresh**. Select MyWirelessNetwork and click **Connect**.
- Enter the Passphrase TSEC@2026
- ***Repeat this for the second Laptop and the Smartphone (under Config > Wireless)***.

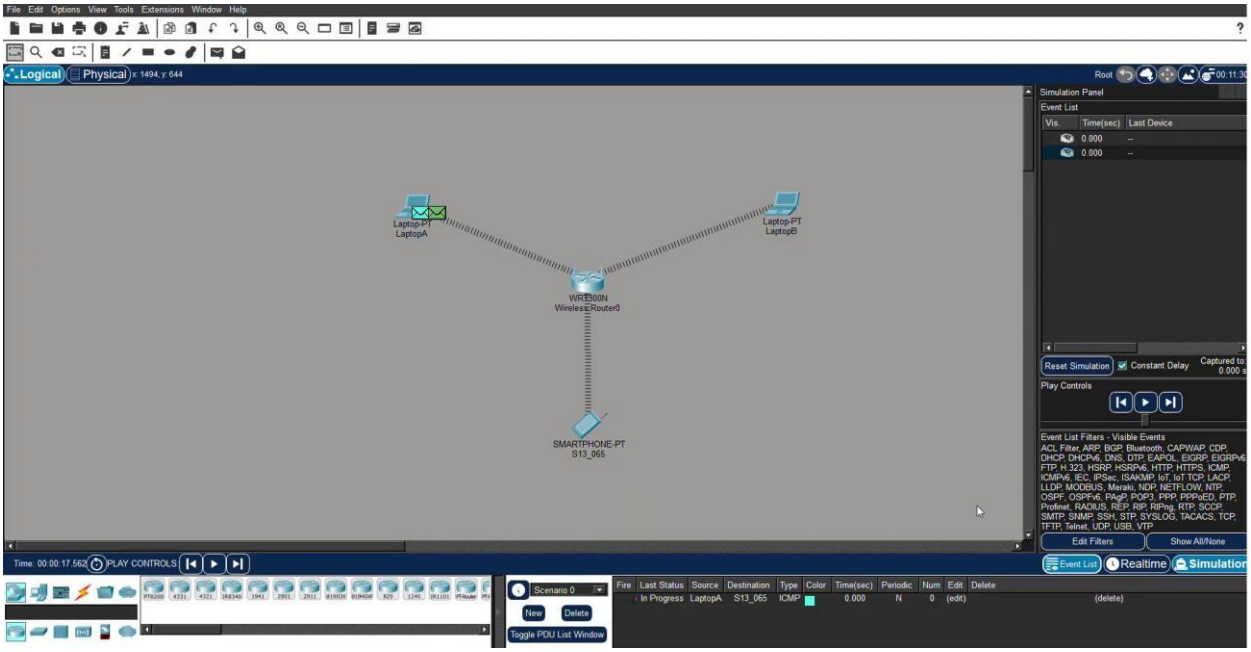
Step 5: Logical Addressing (DHCP Verification)

- Unlike the wired experiment, we will not set Static IPs.
- Click **Laptop** > **Desktop** > **IP Configuration**. Ensure **DHCP** is selected.

Device	**SSID**	**Authenticat**	**Assigned IP Address**
Laptop A	MyWirelessNetwork	WPA2-PSK	192.168.0.101
Laptop B	MyWirelessNetwork	WPA2-PSK	192.168.0.102
Smartphone	MyWirelessNetwork	WPA2-PSK	192.168.0.100

Step 6: Testing Signal Transmission

- **Real-time:**
  - Open the Command Prompt on Laptop A and ping the IP address of the Smartphone.
- **Simulation Mode:**
  - Filter for **ICMP**.
  - Use the Add Simple PDU (Envelope) to click Laptop A then Smartphone. ○ Watch the packet travel through the air (wireless media) to the Router and then to the target device.



Observations

- **No Physical Cables:** Communication was established without the use of Copper or Fiber cables, proving the use of Radio Frequency (RF) as the transmission media.
- **SSID Filtering:** Devices could only see and connect to the specific network identified by the SSID.
- **DHCP Automation:** The Wireless Router automatically managed the IP addresses, simplifying the setup of a WLAN.

Conclusion

The experiment successfully demonstrates signal transmission through **Wireless Media**. It proves that a **Wireless Router** acts as an Access Point and a DHCP server, allowing mobile and portable devices to communicate within a defined range using radio waves and security protocols.